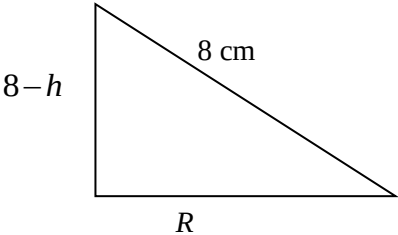


ANSWER SCHEME (MUST 'A' MATHS)

NO.	SOLUTION	MARKS	REMARKS
SECTION A			
1(a)	$\lim_{x \rightarrow 3^-} \frac{ 4x-12 }{x-3}$ $= \lim_{x \rightarrow 3^-} \frac{-(4x-12)}{x-3}$ $= \lim_{x \rightarrow 3^-} \frac{-4(x-3)}{x-3} = -4$	K1 K1J1	Choose the correct function Simplify and answer
1(b)	$\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{(3x + 2)^2}$ $= \lim_{x \rightarrow \infty} \frac{\frac{x^2}{x^2} + \frac{x}{x^2} + \frac{1}{x^2}}{\frac{9x^2}{x^2} + \frac{12x}{x^2} + \frac{4}{x^2}}$ $= \lim_{x \rightarrow \infty} \frac{1 + \frac{1}{x} + \frac{1}{x^2}}{9 + \frac{12}{x} + \frac{4}{x^2}} = \frac{1+0+0}{9+0+0} = \frac{1}{9}$	K1 K1J1	Divide by the highest power of x from the denominator Simplify and answer
1(c)	$\lim_{x \rightarrow 8} \frac{8-x}{\sqrt{6}-\sqrt{x-2}} \left(\frac{\sqrt{6}+\sqrt{x-2}}{\sqrt{6}+\sqrt{x-2}} \right)$ $= \lim_{x \rightarrow 8} \frac{(8-x)(\sqrt{6}+\sqrt{x-2})}{6-(x-2)}$ $= \lim_{x \rightarrow 8} \frac{(8-x)(\sqrt{6}+\sqrt{x-2})}{(8-x)} = 2\sqrt{6}$	K1 K1J1	Multiply with conjugate Simplify and answer
Total		9 marks	
2(a)	$y = 2 - x^2$ $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{2 - (x+h)^2 - (2 - x^2)}{h}$ $= \lim_{h \rightarrow 0} \frac{2 - x^2 - 2xh - h^2 - 2 + x^2}{h}$ $= \lim_{h \rightarrow 0} \frac{h(-2x - h)}{h}$ $= \lim_{h \rightarrow 0} (-2x - h)$ $= -2x$ $\left. \frac{dy}{dx} \right _{x=2} = -2(2) = -4$	K1 K1 J1 J1	Substitute into correct formula Expand & simplify
Total		4 marks	

ANSWER SCHEME (MUST 'A' MATHS)

NO.	SOLUTION	MARKS	REMARKS
SECTION A			
2(b)	$k(x) = \ln\left(\frac{1 - \cos x}{\sin x}\right)$ $= \ln(1 - \cos x) - \ln(\sin x)$ $k'(x) = \frac{\sin x}{1 - \cos x} - \frac{\cos x}{\sin x}$ $= \frac{\sin^2 x - \cos x + \cos^2 x}{(\sin x)(1 - \cos x)}$ $= \frac{1 - \cos x}{(\sin x)(1 - \cos x)}$ $= \frac{1}{\sin x}$ $= \operatorname{cosec} x$	B1 K1 K1 K1 J1	Use logarithm Differentiation Attempt to solve Using identity
Total		5 marks	
3	 $R^2 = 8^2 - (8 - h)^2$ $R^2 = 64 - 64 + 16h - h^2$ $R^2 = 16h - h^2$ $A = \pi R^2$ $A = \pi(16h - h^2)$ $\frac{dA}{dh} = \pi(16 - 2h)$ $h = 6$ $\frac{dA}{dh} = \pi(16 - 2(6))$ $= 4\pi$	K1 K1 J1 K1 K1	Relate R in terms of h Express R^2 in terms of h As A in terms of h Correct differentiation of $\frac{dA}{dh}$ Evaluate $\frac{dA}{dh}$

ANSWER SCHEME (MUST 'A' MATHS)

NO.	SOLUTION	MARKS	REMARKS
SECTION A			
	$\frac{dA}{dt} = \frac{dA}{dh} \times \frac{dh}{dt}$ $= 4\pi(0.2)$ $= 0.8\pi\text{cm}^2\text{s}^{-1}$	K1 J1	Apply Chain Rule
Total		7 marks	

NO.	SOLUTION	MARKS	REMARKS
SECTION B			
1	$(m+ni)(1+i) = 1+2i$ $(a+bi+ni)(1+i) = 1+2i$ $a+bi+ni+ai-b-n = 1+2i$ $a-b-n+bi+ni+ai = 1+2i$ $a-b-n=1 \rightarrow (1) \quad \left. \begin{array}{l} 2a=3 \\ a+b+n=2 \rightarrow (2) \end{array} \right\} a = \frac{3}{2}$ $\sqrt{a^2+b^2} = \sqrt{3}$ $b^2 = 3 - \frac{9}{4} = \frac{3}{4}$ $b = \pm \frac{\sqrt{3}}{2}$ $\frac{3}{2} + \frac{\sqrt{3}}{2} + n = 2 \rightarrow n = 2 - \frac{3}{2} - \frac{\sqrt{3}}{2}$ $\frac{3}{2} - \frac{\sqrt{3}}{2} + n = 2 \rightarrow n = 2 - \frac{3}{2} + \frac{\sqrt{3}}{2}$ $n = \frac{1-\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}$ $m = \frac{3}{2} + \frac{\sqrt{3}}{2}i, \frac{3}{2} - \frac{\sqrt{3}}{2}i$	K1 K1 K1 J1 J1	Substitute $m = a + bi$ Get two equations Simultaneous equation Attempt to find a, b or n . Use formula of modulus for complex number. Values of n Complex number m
Total		5 marks	

ANSWER SCHEME (MUST 'A' MATHS)

NO.	SOLUTION	MARKS	REMARKS																
SECTION B																			
2 (a)	$\frac{y+1}{y} \leq 2$ $\frac{y+1}{y} - \frac{2y}{y} \leq 0$ $\frac{1-y}{y} \leq 0$ <table border="1" data-bbox="279 571 992 795"> <tr> <td></td><td>$(-\infty, 0)$</td><td>$(0, 1]$</td><td>$[1, \infty)$</td></tr> <tr> <td>$1-y$</td><td>+</td><td>+</td><td>-</td></tr> <tr> <td>y</td><td>-</td><td>+</td><td>+</td></tr> <tr> <td>$\frac{1-y}{y}$</td><td>-</td><td>+</td><td>-</td></tr> </table> $\therefore (-\infty, 0) \cup [1, \infty)$ $\log_2 x < 0 \quad \cup \quad \log_2 x \geq 1$ $x < 1 \quad \quad \quad x \geq 2$ $\therefore \{x : 0 < x < 1 \cup x \geq 2\}$		$(-\infty, 0)$	$(0, 1]$	$[1, \infty)$	$1-y$	+	+	-	y	-	+	+	$\frac{1-y}{y}$	-	+	-	K1 J1 K1 J1 K1 K1 J1	Right side 0 Single fraction Table of sign (must have sign) Answer Substitute $\log_2 x$ Solve for both Must in solution set
	$(-\infty, 0)$	$(0, 1]$	$[1, \infty)$																
$1-y$	+	+	-																
y	-	+	+																
$\frac{1-y}{y}$	-	+	-																
Total		7 marks																	
2(b)	$ 1 - 2e^{-x} = e^x$ $\frac{2}{e^x} - 1 = e^x \quad (1 - 2e^{-x} < 0)$ $2 - e^x = e^{2x}$ $e^{2x} + e^x - 2 = 0$ $(e^x + 2)(e^x - 1) = 0$ $e^x + 2 = 0, \quad e^x - 1 = 0$ $e^x = -2 \quad \quad \quad e^x = 1$ $\text{no solution.} \quad \quad \quad x = \ln 1 = 0.$	K1 K1 J1 J1	Form equation with correct modulus definition Get the quadratic equation Get two factors Final Answer																
Total		4 marks																	
3(a)	$f(x) = e^{k+2x}, \quad g(x) = \ln \sqrt{2x}$ $(g \circ g^{-1})(x) = x$ $\frac{1}{2} \ln 2g^{-1}(x) = x$ $\ln 2g^{-1}(x) = 2x \Rightarrow g^{-1}(x) = \frac{e^{2x}}{2}$	K1 J1	Find $g^{-1}(x)$																

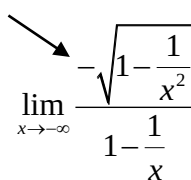
ANSWER SCHEME (MUST 'A' MATHS)

NO.	SOLUTION	MARKS	REMARKS
SECTION B			
3(b)	$(g \circ g^{-1})(3) = g\left[\frac{e^6}{2}\right]$ $= \frac{1}{2} \ln\left(2\left(\frac{e^6}{2}\right)\right)$ $= \frac{1}{2} \ln e^6 = 3$ $(f \circ g)(x) = 2x$ $f[\ln \sqrt{2x}] = 2x$ $e^{k+2[\ln \sqrt{2x}]} = 2x$ $e^{k+\ln 2x} = 2x$ $k + \ln 2x = \ln 2x$ $k = 0$	K1 J1 K1 K1 K1 J1	Substitute $g^{-1}(3)$ Substitute $g(x)$ Substitute $f(x)$ Attempt to solve
Total		8 marks	
4	$g(x) = 2 - \frac{5}{x}, x \neq 0.$ (a) Domain $g(x): \mathbb{R} \setminus \{0\}$. (b) $g_1(x) = g_2(x)$ $2 - \frac{5}{x_1} = 2 - \frac{5}{x_2}$ $\frac{5}{x_1} = \frac{5}{x_2}$ $x_1 = x_2$ $g(x)$ is a one-to-one function. Therefore, $g^{-1}(x)$ exist. (c) $(h \circ g)(x) = h\left(2 - \frac{5}{x}\right) \quad (g \circ h)(x) = g\left(\frac{5}{2-x}\right)$ $= \frac{5}{2 - \left(2 - \frac{5}{x}\right)} \quad = 2 - \frac{5}{\frac{5}{2-x}}$ $= \frac{5}{\frac{5}{x}} \quad = 2 - (2-x)$ $= x \quad = x$	B1 K1 J1 K1 K1 J1	Let $g(x_1) = g(x_2)$ Get $x_1 = x_2$ One to one and inverse exist. Find both $(h \circ g)(x)$ and $(g \circ f)(x)$ Solve composite functions Get $(h \circ g)(x) = x$ and

ANSWER SCHEME (MUST 'A' MATHS)

NO.	SOLUTION	MARKS	REMARKS
SECTION B			
(d)	Since $(h \circ g)(x) = (g \circ h)(x) = x$, $g(x)$ and $h(x)$ are inverse to each other Domain $h(x) : \mathbb{R} \setminus \{2\}$. Range $h(x) = \text{Domain } g(x) : \mathbb{R} \setminus \{0\}$. Graph of $g(x)$ and $g^{-1}(x)$.	J1 B1 B1 R1 R1 R1	$(g \circ h)(x) = x$ Conclusion Pattern reciprocal function, $g(x)$ At least one side correct for inverse function, $g^{-1}(x)$ All correct (including interception with axes) *if no label on function, R0 R0 R0
Total		12 marks	

ANSWER SCHEME (MUST 'A' MATHS)

NO.	SOLUTION	MARKS	REMARKS
SECTION B			
5(a)	$f(x) = \frac{\sqrt{x^2 - 1}}{x - 1}$ $D_f = (-\infty, -1] \cup (1, \infty)$ $\lim_{x \rightarrow 1^+} \left(\frac{\sqrt{x^2 - 1}}{x - 1} \right) \left(\frac{\sqrt{x^2 - 1}}{\sqrt{x^2 - 1}} \right)$ $= \lim_{x \rightarrow 1^+} \frac{x^2 - 1}{(x - 1)\sqrt{x^2 - 1}}$ $= \lim_{x \rightarrow 1^+} \frac{(x - 1)(x + 1)}{(x - 1)\sqrt{x^2 - 1}}$ $= \lim_{x \rightarrow 1^+} \frac{(x + 1)}{\sqrt{x^2 - 1}} = +\infty$ $\therefore x = 1 \text{ is a vertical asymptote}$ $\lim_{x \rightarrow +\infty} \frac{\sqrt{x^2 - 1}}{x - 1}$ $= \lim_{x \rightarrow +\infty} \frac{\sqrt{\frac{x^2}{x^2} - \frac{1}{x^2}}}{\frac{x}{x} - \frac{1}{x}}$ $= \lim_{x \rightarrow +\infty} \frac{\sqrt{1 - \frac{1}{x^2}}}{1 - \frac{1}{x}} = 1$ $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 - 1}}{x - 1}$ $= \lim_{x \rightarrow -\infty} \frac{-\sqrt{\frac{x^2}{x^2} - \frac{1}{x^2}}}{\frac{x}{x} - \frac{1}{x}}$ $= \lim_{x \rightarrow -\infty} \frac{-\sqrt{1 - \frac{1}{x^2}}}{1 - \frac{1}{x}} = -1$ $\therefore y = 1 \text{ and } y = -1 \text{ are horizontal asymptotes}$	K1 J1 K1 K1 J1	Must 1^+ Both $+\infty$ and conclusion Both divide by the highest degree of the denominator  Both correct
Total		5 marks	
5(b)	$\lim_{x \rightarrow 1^+} \frac{ 1 - x }{x - 1} + b = \lim_{x \rightarrow 1^-} \frac{x^2 - 1}{x - 1}$ $ 1 - x = \begin{cases} 1 - x, & x < 1 \\ x - 1, & x > 1 \end{cases}$ $\lim_{x \rightarrow 1^+} \frac{x - 1}{x - 1} + b = \lim_{x \rightarrow 1^-} \frac{(x - 1)(x + 1)}{x - 1}$ $1 + b = 2$ $b = 1$	K1J1 K1K1 K1 J1	Equal limit with correct function left and right of 1 K1 Choose correct function for 1^+ K1 Factorise for 1^- K1 Simplify

ANSWER SCHEME (MUST 'A' MATHS)

NO.	SOLUTION	MARKS	REMARKS
SECTION B			
	$f(1) = \lim_{x \rightarrow 1} f(x)$ $a^2 - 7 = 2$ $a^2 = 9$ $a = \pm 3$	K1K1 J1	K1 $f(1) = \lim_{x \rightarrow 1} f(x)$ K1 his $\lim_{x \rightarrow 1} f(x)$
Total		9 marks	
6(a)	$\ln(3y) = y \ln x$ $\frac{1}{y} \frac{dy}{dx} = y \left(\frac{1}{x} \right) + (\ln x) \frac{dy}{dx}$ $\left(\frac{1}{y} - \ln x \right) \frac{dy}{dx} = \frac{y}{x}$ $\left(\frac{1 - y \ln x}{y} \right) \frac{dy}{dx} = \frac{y}{x}$ $\frac{dy}{dx} = \frac{y^2}{x(1 - y \ln x)}$ Hence, when $x = 1$, $\ln 3y = y \ln 1$ $\ln 3y = 0$ $3y = e^0$ $y = \frac{1}{3}$ $\left. \frac{dy}{dx} \right _{\left(1, \frac{1}{3}\right)} = \frac{\left(\frac{1}{3}\right)^2}{1 \left(1 - \frac{1}{3} \ln 1\right)} = \frac{1}{9}$	K1K1 K1 J1 K1 K1 J1	Differentiate LHS & RHS Simplify Find the value of y
Total		7 marks	
6(b)	$x = \cos 2t \qquad y = \sin^2 t + 2e^{\ln t} = \sin^2 t + 2t$ $\frac{dx}{dt} = -2 \sin 2t \qquad , \qquad \frac{dy}{dt} = 2 \sin t \cos t + 2$	K1 K1	Find $\frac{dx}{dt}$ and $\frac{dy}{dt}$

ANSWER SCHEME (MUST 'A' MATHS)

NO.	SOLUTION	MARKS	REMARKS												
SECTION B															
	$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$ $= (2 \sin t \cos t + 2) \left(\frac{1}{-2 \sin 2t} \right)$ $= \frac{\sin 2t + 2}{-2 \sin 2t}$ $= \frac{\sin 2t}{-2 \sin 2t} + \frac{2}{-2 \sin 2t}$ $= -\frac{1}{2} - \operatorname{cosec} 2t \quad (\text{shown})$ $\frac{d^2 y}{dx^2} = 2 \operatorname{cosec} 2t \cot 2t \frac{dt}{dx}$ $= (2 \operatorname{cosec} 2t \cot 2t) \left(\frac{1}{-2 \sin 2t} \right)$ $= \frac{\operatorname{cosec} 2t \cot 2t}{-\sin 2t}$ $\frac{d^2 y}{dx^2} = \frac{\operatorname{cosec} 2 \left(\frac{\pi}{8} \right) \cot 2 \left(\frac{\pi}{8} \right)}{-\sin 2 \left(\frac{\pi}{8} \right)}$ $= \frac{\sqrt{2}(1)}{-\frac{1}{\sqrt{2}}}$ $= -2$	K1 K1 J1 K1 K1 K1 J1	Substitute into chain rule Use identity & simplify Correct differentiation Substitute $\frac{dt}{dx}$ Substitute value into his $\frac{d^2 y}{dx^2}$												
Total		9 marks													
7	$f(x) = x^3 + 3x^2 - 9x + 6$ $f'(x) = 3x^2 + 6x - 9$ $0 = 3(x+3)(x-1)$ $x = -3, x = 1$ <table border="1" data-bbox="279 1780 970 1960"> <tr> <td>Interval</td><td>$(-\infty, -3)$</td><td>$(-3, 1)$</td><td>$(1, \infty)$</td></tr> <tr> <td>Sign of $\frac{dy}{dx}$</td><td>+</td><td>-</td><td>+</td></tr> <tr> <td>slope</td><td>increasing</td><td>decreasing</td><td>Increasing</td></tr> </table>	Interval	$(-\infty, -3)$	$(-3, 1)$	$(1, \infty)$	Sign of $\frac{dy}{dx}$	+	-	+	slope	increasing	decreasing	Increasing	K1 J1 K1	Find the first derivative Table of sign
Interval	$(-\infty, -3)$	$(-3, 1)$	$(1, \infty)$												
Sign of $\frac{dy}{dx}$	+	-	+												
slope	increasing	decreasing	Increasing												

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ANSWER SCHEME (MUST 'A' MATHS)

NO.	SOLUTION	MARKS	REMARKS
SECTION B			
Total		9 marks	